

Ramakrishnan Sundara Raman

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🔗 [Personal Website](#)

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Positions

- **University of California Santa Cruz**
Department of Computer Science and Engineering, Baskin School of Engineering
Assistant Professor Jul 2024 – Present
- **Cloudflare**, *Research Intern* Jul 2022 – Sep 2022
- **The Citizen Lab**, *OTF Senior ICFP Fellow* Nov 2021 – Jul 2022
- **Microsoft**, *Associate Consultant* Jul 2017 – May 2018
- **University of Maryland**, *Research Intern* Mar 2017 – Jul 2017

Education

- 2018 – 2024 📖 **Ph.D. in Computer Science, University of Michigan** in Computer Science and Engineering
Advisor: Prof. Roya Ensafi
- 2013 – 2017 📖 **Bachelor of Technology, Vellore Institute of Technology** in Computer Science and Engineering
Project Title: Measuring Last-Mile Internet Reliability During Severe Weather

Conference Publications

- **CenRL: A Framework for Performing Intelligent Censorship Measurements** 🔗
H. Le, A. Huremagic, K. Wang, R. Ensafi, **R. Sundara Raman**
IEEE Security and Privacy (S&P), May 2026
- **Fingerprinting Deep Packet Inspection Devices by their Ambiguities** 🔗
D. Xue, A. Huremagic, W. Wang, **R. Sundara Raman**, R. Ensafi
ACM SIGSAC Conference on Computer and Communications Security (CCS), October 2025
🏆 IRTF Applied Networking Research Prize
- **Digital Discrimination of Users in Sanctioned States: The Case of the Cuba Embargo** 🔗
A. Ablove, S. Chandrashekar, H. Le, **R. Sundara Raman**, R. Ramesh, H. Oppenheimer, R. Ensafi
USENIX Security Symposium, August 2024
🏆 Distinguished Paper Award
- **Modeling and Detecting Internet Censorship Events** 🔗
E. Tsai, **R. Sundara Raman**, A. Prakash, R. Ensafi
Network and Distributed System Security Symposium (NDSS), February 2024
- **Global, Passive Detection of Connection Tampering** 🔗
R. Sundara Raman, L.H. Merino, K. Bock, M. Fayed, D. Levin, N. Sullivan, L. Valenta
ACM SIGCOMM, September 2023
- **Network Responses to Russia's Invasion of Ukraine in 2022: A Cautionary Tale for Internet Freedom** 🔗
R. Ramesh*, **R. Sundara Raman***, A. Virkud, A. Dirksen, A. Huremagic, D. Fifield, D. Rodenburg, R. Hynes, D. Madory, R. Ensafi
USENIX Security Symposium, August 2023
🏆 FOCI Best Practical Award 2024
- **CERTainty: Detecting DNS Manipulation using TLS Certificates** 🔗
E. Tsai, D. Kumar, **R. Sundara Raman**, G. Li, Y. Eiger, R. Ensafi
Privacy Enhancing Technologies Symposium (PETS), July 2023
- **Network Measurement Methods for Locating and Examining Censorship Devices** 🔗
R. Sundara Raman*, M. Wang*, J. Dalek, J. Mayer, and R. Ensafi

*Both authors contributed equally to this research

- **A Large-scale Investigation into Geodifferences in Mobile Apps** 🔗
R. Kumar, A. Virkud, **R. Sundara Raman**, A. Prakash, and R. Ensafi
USENIX Security Symposium, August 2022
- **Lost in Transmission: Investigating Filtering of COVID-19 Websites** 🔗
A. Vyas, **R. Sundara Raman**, N. Ceccio, P. Lutscher, and R. Ensafi
Financial Cryptography and Data Security (FC), March 2021
- **Censored Planet: An Internet-wide, Longitudinal Censorship Observatory** 🔗
R. Sundara Raman, P. Shenoy, K. Kohls, and R. Ensafi
ACM SIGSAC Conference on Computer and Communications Security (CCS), November 2020
- **Investigating Large Scale HTTPS Interception in Kazakhstan** 🔗
R. Sundara Raman, L. Evdokimov, E. Wustrow, A. Halderman, and R. Ensafi
ACM Internet Measurement Conference (IMC), October 2020
🏆 Best Paper Award Nominee
- **Measuring the Deployment of Network Censorship Filters at Global Scale** 🔗
R. Sundara Raman, A. Stoll, J. Dalek, R. Ramesh, W. Scott, R. Ensafi
Network and Distributed System Security Symposium (NDSS), February 2020
- **Decentralized Control: A Case Study of Russia** 🔗
R. Ramesh, **R. Sundara Raman**, M. Bernhard, V. Ongkowijaya, L. Evdokimov, A. Edmundson, S. Sprecher, M. Ikram, R. Ensafi
Network and Distributed System Security Symposium (NDSS), February 2020
🏆 CSAW'20 Applied Research Competition North America Top 10 Paper

Workshop Publications

- **From Underground to Mainstream Marketplaces: Measuring AI-Enabled NSFW Deepfakes on Fiverr** 🔗
M. M. Dawoud, A. Cuevas, **R. Sundara Raman**
Symposium on Usable Security and Privacy (USEC), February 2026
- **Not What It Used To Be: Generational Analysis of Top-level Domain Reputation**
J. Szurdi, R. Ramesh, **R. Sundara Raman**, D. Liu
Workshop on Measurements, Attacks, and Defenses for the Web (MADWeb) 2026
🏆 Distinguished Paper Award
- **Advancing the Art of Censorship Data Analysis** 🔗
R. Sundara Raman, A. Virkud, S. Laplante, V. Fortuna, R. Ensafi
Free and Open Communications on the Internet (FOCI), February 2023
- **A Multi-Perspective View of Internet Censorship in Myanmar** 🔗
R. Padmanabhan, A. Filastò, M. Xynou, **R. Sundara Raman**, K. Middleton, M. Zhang, D. Madory, M. Roberts, and A. Dainotti
Free and Open Communications on the Internet (FOCI), August 2021

Awards

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| Research | ■ MADWeb 2026 Best Paper Award
February 2026 |
| | ■ IRTF Applied Networking Research Prize (ANRP) 2026
January 2026 |
| | ■ USENIX Security 2024 Distinguished Paper Award 🔗
August 2024 |
| | ■ FOCI'24 Best Practical Award 🔗
July 2024 |

	■ FOCI'23 Rising Star Award 🔗 July 2023 ■ IRTF Applied Networking Research Prize (ANRP) 2023 🔗 January 2023 ■ University of Michigan CSE Graduate Student Honors Runners Up Award 🔗 November 2020 ■ ACM IMC 2020 Best Paper Award Nominee October 2020
Fellowships	■ Rackham Predoctoral Fellowship 🔗 March 2023 ■ OTF Information Controls Fellowship (ICFP) 🔗 November 2021
Teaching & Service	■ Travel Grants–ACM IMC 2022, USENIX Security 2020, and Citizen Lab Summer Institute 2019 ■ North Campus Deans' MLK Spirit Award 2023 🔗 January 2023 ■ University of Michigan CSE GSI Honorable Mention Award 🔗 August 2021
Student Awards	■ Rohit Nagubandi, Tor SEEKCommons Fellowship 🔗 February 2025 ■ Sriya Katreddi, Deans Award for Undergraduate Research 🔗 June 2025 ■ Prasiddh Pooskur, Deans Award for Undergraduate Research 🔗 June 2025

Mentoring

- **PhD Students:** Linden Chen (2025-Present), Mohamed Moustafa Dawoud (2025-Present), Ramon Torres (2025-Present)
- **Masters Students:** Kevin Tai (2026-Present), Rohit Nagubandi (2024-2026), Riya Aggarwal (2025), Likith Rahul Krishnamurthy (2025), Kevin Wang (Michigan, 2023)
- **Undergraduate Students:** Nate Jarett (2026-Present), Ethan Cao (2026-Present), Ruthwika Gajjala (2026-Present), Shubham Marathe (2026-Present), Rachit Verma (2026-Present), Isa Abello (2025-Present), Andrew Chu (2025-Present), Geoffrey Tang (2025-Present), Graham Klinger (2025-Present), Harshita Sarathy (2025-Present), Mukesh Saravanan (2025-Present), Kenneth Lai (2025-Present), Sriya Katreddi (2024-Present), Peter Dobbins (2025), Michael Kamensky (2025), Dominick Rangel (2024-2025), Felicity Shih (2024-2025), Carlo Aqui (2025), Leonardo Gallego (2024), Brian Huang (2024-2025), Prasiddh Pooskur (2024-2025), Ryan Sangha (2024-2025), Nadav Oren (Michigan, 2023), Apurva Virkud (Michigan, 2020-2022), Anjali Vyas (Michigan, 2020-2021), Prerana Shenoy (Michigan, 2019-2020)

Teaching

Winter 2025, 2026	■ Instructor, CSE 153, UC Santa Cruz Topic: Network Security and Privacy (Undergraduate)
Fall 2024, 2025	■ Instructor, CSE 253, UC Santa Cruz Topic: Network Security (Graduate)
Spring 2025	■ Instructor, CSE 156, UC Santa Cruz Topic: Network Programming (Undergraduate)
March 2023	■ Guest Lecturer, CS 4973, Northeastern University Topic: Censorship and Circumvention
January 2023	■ Substitute Instructor, EECS 588, University of Michigan Topic: Computer and Network Security
March 2022	■ Guest Lecturer, EECS 588, University of Michigan Topic: Censorship and Circumvention

Talks

- **Invited Talk: Advancing the Art of Internet Censorship Measurement**
Dartmouth College, October 2025
- **Invited Talk: Network Measurement Methods for Locating and Examining Network Censorship Devices** 
IETF 118, November 2023
- **Conference Talk: Global, Passive Detection of Connection Tampering** 
ACM SIGCOMM, September 2023
- **Conference Talk: Network Responses to Russia's Invasion of Ukraine in 2022: A Cautionary Tale for Internet Freedom**
Usenix Security Symposium, August 2023
- **Workshop Talk: Advancing the Art of Censorship Data Analysis**
Free and Open Communications on the Internet (FOCI), February 2023
- **Lightning Talk: Exploring Internet censorship through the Censored Planet Dashboard**
Open Technology Fund (OTF), January 2023
- **Conference Talk: Network Measurement Methods for Locating and Examining Network Censorship Devices**
Conference on emerging Networking EXperiments and Technologies (CoNEXT 2022), December 2022
- **Tech Demo: Exploring Internet censorship with the Censored Planet Dashboard** 
RightsCon 2022, June 2022
- **Symposium Talk: Censored Planet: An Internet-wide, Longitudinal Censorship Observatory**
The 5th Annual Data for Public Good Symposium, March 2022
- **Invited Talk: Censored Planet Community Webinar 2021** 
Censored Planet, October 2021
- **Invited Talk: Censored Planet: An Internet-wide Longitudinal Observatory**
NDSS DNS Privacy Workshop (DNSPriv), February 2021
- **Conference Talk: Censored Planet: An Internet-wide Longitudinal Observatory** 
ACM Conference on Computer and Communications Security (CCS), November 2020
- **Conference Talk: Investigating Large Scale HTTPS Interception in Kazakhstan** 
ACM Internet Measurement Conference (IMC), October 2020
- **Invited Talk: Censored Planet: Measuring Internet censorship remotely** 
OONI Internet Measurement Village (IMV), June 2020
- **Conference Talk: Measuring the Deployment of Network Censorship Filters at Global Scale** 
Network and Distributed System Security Symposium (NDSS), February 2020

Reports and Blogposts

- **Insights from data at large scale might just be a small miracle**
The Cloudflare Blog, October 2025
 <https://blog.cloudflare.com/experience-of-data-at-scale/#insights-from-data-at-large-scale-might-just-be-a-small-miracle>
- **A Global Assessment of Third-Party Connection Tampering**
The Cloudflare Blog, September 2024
 <https://blog.cloudflare.com/connection-tampering/>
- **Measuring Connection Tampering Around the World**

Internet Society Pulse Blog, October 2023

<https://pulse.internetsociety.org/blog/measuring-connection-tampering-around-the-world>

○ **Making Censorship Data More User Friendly**

Internet Society Pulse Blog, May 2023

<https://pulse.internetsociety.org/blog/making-censorship-data-more-user-friendly>

○ **Profiling Internet Censors: Methods to Measure and Locate Censorship Devices**

Open Technology Fund (OTF), December 2022

<https://www.opentech.fund/news/profiling-internet-censors-methods-to-measure-and-locate-censorship-devices/>

○ **An analysis of a large-scale HTTPS interception**

APNIC Blog, February 2021

<https://blog.apnic.net/2021/02/15/an-analysis-of-a-large-scale-https-interception/>

○ **US Government and military websites are geoblocked from Hong Kong and China**

Censored Planet, August 2020

<https://censoredplanet.org/hongkong>

Service

Chairing	■ Session Chair at ACM Internet Measurement Conference (IMC) 2025 🔗
	■ Session Chair at Free and Open Communications on the Internet (FOCI) 2021 🔗
Reviewing	■ TPC Member, ACM SICOMM Education Workshop 2026 🔗
	■ TPC Member, ACM Conference on Computer and Communications Security (CCS) 2026 🔗
	■ TPC Member, ACM Internet Measurement Conference (IMC) 2026 🔗
	■ Reviewer, IEEE/ACM Transactions on Networking (TON) 2025 🔗
	■ TPC Member, ACM Internet Measurement Conference (IMC) 2025 🔗
	■ TPC Member, Free and Open Communications on the Internet (FOCI) 2025 🔗
	■ TPC Member, IEEE Symposium on Security and Privacy (S&P) 2025 🔗
	■ TPC Member, ACM Internet Measurement Conference (IMC) 2024 🔗
	■ TPC Member, IEEE Symposium on Security and Privacy (S&P) 2024 🔗
	■ TPC Member, Free and Open Communications on the Internet (FOCI) 2024 🔗
	■ Reviewer, 6th International Conference on Computer Science and Application Engineering (CSAE) 2022
Leadership	■ TPC Member, ACM SIGCOMM 2021 Workshop on Free and Open Communications on the Internet (FOCI) 🔗
	■ Security Seminar Host, UCSC September 2024 – Present
	■ Outreach Chair at CSEG, UM CSE graduate student organization June 2020 – May 2022 My responsibilities included arranging and managing outreach events throughout the year that aimed at spreading knowledge about computer science research in various schools, colleges and groups. 🏆 Deans' MLK Spirit Award 2023
	■ Administrator of the Security Reading Group (SECRIT) in UM CSE June 2019 – June 2021